

Design of Local Area DGNSS Architecture to Support Unmanned Aerial Vehicle Networks: Optimal Integrity/Continuity Allocations and Fault Monitoring

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ABSTRACT

A local-area differential global navigation satellite system (LAD-GNSS) provides high integrity and accuracy for unmanned aerial vehicles (UAVs). In this paper, we propose modifications to improve the LAD-GNSS performance and integrity using a two-way datalink and fully analyze the performance of the system. Key components of the proposed system include an operation requirement design, modification of integrity monitoring, and optimal allocation of integrity/continuity budgets.

Two operation requirements, vertical alert limit (VAL) and tolerable error limit (TEL), are derived based on the operating environment and uncertainty of navigation. The vertical protection level (VPL) under the optimal allocation and maximum ionospheric error in vertical, computed in real time, are, respectively, compared to VAL and TEL to determine if the system meets the integrity requirements for a given operation. In addition, a method is described for deriving PLs for excessive acceleration and code-carrier divergence fault scenarios, which are bounded by using a maximum-allowable error in range. Using these PLs, the integrity/continuity allocations are ideally and dynamically assigned to each single-fault hypothesis to obtain optimized PLs that are identical for all fault scenarios. The results show that the VPLs are reduced by approximately 8% when implementing the optimal allocation method.

1. INTRODUCTION

One of the most essential technologies required for the operation of unmanned aerial vehicles (UAVs) is reliable and accurate computation of the navigation solution. The concept of a local-area UAV network that utilizes local-area differential global navigation satellite system (LAD-GNSS) solutions was proposed to achieve high navigation accuracy and reliability, which are required for autonomous UAV applications [1]. LAD-GNSS meets the required level of integrity requirements (comparable to those of GBAS Category I–III operations) by monitoring navigation faults at both the ground station and UAV and by broadcasting integrity information to the UAV. By utilizing the integrity information, which defines navigation system error models, the UAVs can compute their conservative position error bounds (i.e., protection

levels (PLs)) and consequently safe separation distances in real time. The LAD-GNSS architecture for the local-area UAV network proposed in previous work includes several methods of simplifying the current GBAS integrity monitoring algorithms and hardware configurations to lower the system cost and complexity while maintaining an acceptable safety level [1]. The performance and integrity of LAD-GNSS can also be improved by designing additional modules using an airborne-to-ground datalink.

The LAD-GNSS integrity monitor testbed was installed on the KAIST campus. However, the above-mentioned modifications, including ground hardware/software changes and airborne components, have not yet been fully analyzed and implemented from a system-design perspective.

In this study, several key components of the LAD-GNSS architecture were developed, including system operation coverage and requirements, modifications of integrity monitors in both ground and airborne subsystems, and optimal integrity/continuity allocations. In the current GBAS architecture, separate PL values for three scenarios (i.e., fault-free (H0), single reference-receiver fault (H1), and ephemeris-fault cases) are herein defined based on the predetermined integrity and continuity allocations. These allocations can ideally be dynamically assigned to make the PLs identical for all fault scenarios.

The use of a two-way datalink to relay UAV information back to the ground station provides a significant advantage over the conventional GBAS and enables allocations in real time. Since the LAD-GNSS ground station can receive information about the satellite geometry utilized by the UAV for its positioning, it is possible to compute the PLs for all fault scenarios and optimally allocate the integrity and continuity budgets to make the PLs identical for all fault scenarios.

The LAD-GNSS integrity monitors were developed using the well-established GBAS as a starting point. In this study, we implemented major modifications on the ground subsystems using a two-way datalink. First, by using a known user geometry, the ionospheric geometry screening was simplified to calculate the maximum ionospheric error in vertical (MIEV) for the all-in-view geometry instead of every subset of satellite geometries. MIEV represents the position error due to the undetected ionospheric range error in the presence of the code-carrier divergence (CCD)

monitor and ground dual-smoothing ionosphere gradient monitor algorithm (DSIGMA).

The remainder of this paper is organized as follows. Section 2 illustrates the LAD-GNSS architecture, including the operation coverage and integrity monitoring modifications. Section 3 explains the operation requirement of LAD-GNSS, and Section 4 details the method of optimal integrity and continuity allocations. Section 5 presents an analysis of the geometry screening for mitigating the worst-case ionospheric gradients. In addition, the simulation results are presented. Section 6 concludes the paper.

2. LAD-GNSS ARCHITECTURE FOR UAV

The proposed LAD-GNSS architecture for UAV enables a low-cost space-conserving installation and low complexity on monitoring algorithms for commercial applications, while supporting high navigation integrity within the airspace coverage. Figure 1 illustrates the LAD-GNSS architecture concept for UAV. The ground facility shown at the center is the source of the LAD-GNSS correction and integrity information, as well as real-time guidance for each UAV operating within the coverage.

To enable safe UAV operations in the civil airspace, the coverage by the system should be carefully defined. As the distance from the ground facility to an operating UAV increases, the residual navigation errors that remain increase, even those after applying differential corrections. Thus, the error uncertainty over the distance from the ground facility to UAVs should be considered when defining the coverage to be safely integrated in the civil airspace.

The coverages of the proposed system were pre-defined based on expected UAV missions for each region for choosing the operation radius. They were additionally based on the simulation results for vertical position uncertainty (including the vertical protection level (VPL) and MIEV) for determining the operation heights. The defined coverage airspaces are listed in Table 1.

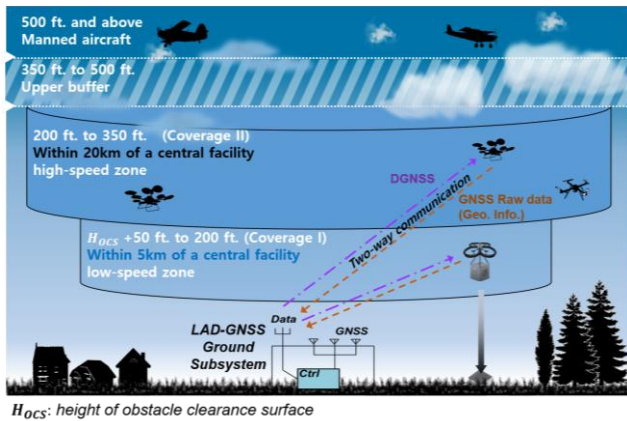


Figure 1. Concept of LAD-GNSS architecture for the UAV

Table 1. LAD-GNSS coverage

Criteria	Radius (from the ground facility)	Height
Coverage I	~ 5 km	$50 \text{ ft} + H_{OCS} \leq h_1 < 200 \text{ ft}$
Coverage II	5 km ~ 20 km	$200 \text{ ft} \leq h_2 < 350 \text{ ft}$

‘Coverage I’ supports low-speed UAV operations from a 200-ft altitude down to an altitude of $50 \text{ ft} + H_{OCS}$ within the radius of 5 km. The minimum altitude, 50 ft, was chosen from the simulations of VPL and MIEV at a distance of 5 km from the ground facility (where the worst-case VPLs and MIEVs were derived). The simulation details are discussed in Sections 4 and 5.

H_{OCS} indicates the maximum height of an obstacle that can be placed along the aircraft approach path. H_{OCS} for GBAS have been defined and are enforced at runways that have precision approaches, non-precision approaches, and even visual operations [2]. H_{OCS} for UAVs should be carefully defined based on the environment above which the UAV is actually flying.

A potential application proposed for Coverage I may be a UAV landing mission down to $50 \text{ ft} + H_{OCS}$. Below the minimum height, the aid of other sensors, such as a vision sensor or laser sensor, is required for touchdown. ‘Coverage II’ was designed to support high-speed operations between 200 ft and 350 ft within the radius of 20 km from the ground station. The upper limit, 350 ft, was determined to avoid penetration of the airspace where manned aircrafts operate (above 500 ft) owing to the vertical position uncertainty. The magnitude of the upper buffer in Figure 1 from 350 ft to 500 ft is defined in the same manner as the lower limit. The only difference is that H_{OCS} is not considered for determining the upper buffer.

A key feature of the proposed system is to utilize a two-way datalink between the ground facility and UAVs in the navigation system that was originally used for the control and guidance function. The proposed system relays GNSS raw measurements received from UAVs to the ground facility in real time, as shown in Figure 2. The ground facility performs monitoring, correction generation, and PL computation. After all processes are performed at the ground facility, UAVs only receive differential corrections and compute their corrected position solution on onboard. PLs and alert messages are transmitted to the ground controller for safety assured guidance. The intent of this proposed approach is to rely on major computations at the ground facility, rather than at the onboard receivers, to reduce the onboard computation burden for ease of implementation. This modified configuration provides significant benefits to fault monitoring algorithms and PL computation algorithms by having known user satellite geometries.

Figure 2 describes the modified configuration and components of LAD-GNSS. Three ground reference receivers are connected to low-cost antennas. Siting constraints for most commercial applications will not have space to spread ground antennas over hundreds of meters.

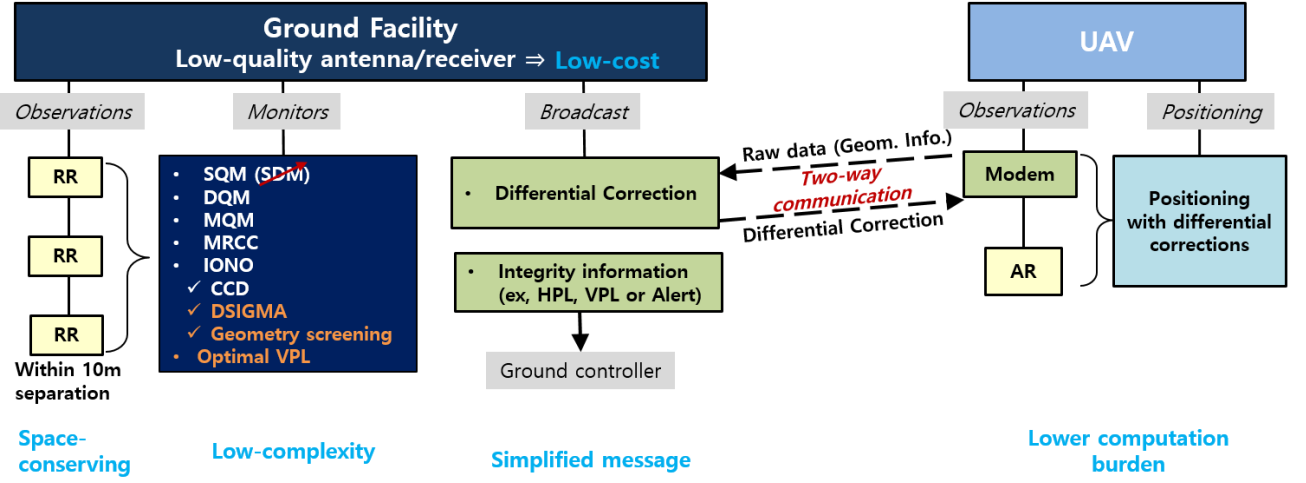


Figure 2. LAD-GNSS architecture configuration and components

In this system, antennas are separated from each other by 10 m or less. Low-quality antennas and a relatively short antenna separation increase the impact of multipath errors. To mitigate performance degradation due to the hardware restriction, and to leverage the new system configuration, we developed several key components of the LAD-GNSS architecture, including optimal integrity allocations and modifications of integrity monitors. The optimal application should reduce PLs compared to the original PLs computed from the pre-allocated integrity budgets. The algorithms for optimal PL computation are introduced in Section 4.

In addition to the optimal allocation of integrity budget, the modification of monitors can potentially improve the overall performance. To mitigate the integrity risk due to anomalous ionosphere gradients, the DSIGMA monitor (adopted for GBAS CAT-II/III) was added with the use of a code-carrier divergence (CCD) monitor to meet the system requirement. Moreover, geometry screening was simplified based on the known user geometry.

The simplified geometry screening monitor operates to compute MIEV in real time. Since there is no need to consider all possible user geometries, both reduced computation load and improved availability would be the resulting benefits of the modified geometry screening. The concept of modified geometry screening and the MIEV simulations are discussed in Section 5. Lastly, the signal deformation monitor (SDM) can be eliminated by selecting ground and airborne GNSS receivers that are either the same or very similar. This would cancel the resulting error when users apply their differential corrections [1].

3. OPERATION REQUIREMENT FOR LAD-GNSS FOR UAV

3.1 Coverage I and II Vertical Alert Limits

For GBAS, PLs are compared to alert limit in each position axis of interest to determine if the system meets the integrity and continuity requirements for a given operation. A VAL of 10 m for GBAS CAT I was established based on the vertical error characteristics of the

instrument landing system (ILS) [3]. Since Coverage II of LAD-GNSS has the same low limit as the GBAS CAT I decision height of 200 ft, the UAV operation in Coverage II uses VAL of 10 m to determine if it meets the system requirements. However, the UAV operating in Coverage I may require a different value from CAT-I VAL because flies as low as 50 ft.

In this paper, the VAL of Coverage I is analyzed in a manner similar to GBAS CAT III. The analysis focuses on the total system error (TSE), specifically its composition of the sum of the navigation sensor error (NSE) and flight technical error (FTE). The key requirement for vertical TSE performance of GBAS CAT III is ensuring a touchdown no less than 200 ft from a threshold, and no more than 2,700 ft [3]. Similar to GABS CAT III, the performance criteria of Coverage I of LAD-GNSS are stated in terms of a UAV height lower than H_{OCS} . Corresponding to these requirements, the analysis is divided into a fault-free NSE case, followed by a faulted NSE case. The allowed risk for the fault-free case is 10^{-6} ; the allowed risk for the faulted case is 10^{-5} [3, 4]. The paper handles only the fault-free NSE case because it has a tighter requirement than that of the faulted case.

To facilitate the analysis, simplified assumptions are used in this analytical derivation. It is assumed that the relationship between vertical NSE and VAL at the worst geometry is expressed as

$$\sigma_{NSE,v}(m) = \frac{VAL(m)}{k_{FFMD}}; \quad k_{FFMD} = 5.81 \quad (1)$$

where k_{FFMD} is a multiplier defined to ensure a specified probability of fault-free missed detection. The vertical TSE is normally distributed with a zero mean and standard deviation ($\sigma_{TSE,v}$) given by

$$\sigma_{TSE,v} = \sqrt{\sigma_{NSE,v}^2 + \sigma_{FTE,v}^2} \quad (2)$$

It is assumed that, if TSE is zero, the UAV height will be 50 ft from H_{OCS} . Therefore, the probability that the height of UAV will be lower than H_{OCS} is given by

$$P_{risk} = \int_{-\infty}^{-50} \frac{1}{\sqrt{2\pi}\sigma_{TSE,v}} e^{-\frac{x^2}{2\sigma_{TSE,v}^2}} dx \quad (3)$$

Figure 3 shows a plot of the fault-free risk of Coverage I versus VAL for values of $\sigma_{FTE} = 3\text{ m}$, $\sigma_{FTE} = 2.7\text{ m}$, $\sigma_{FTE} = 2\text{ m}$, and $\sigma_{FTE} = 1.5\text{ m}$. Note that, for $\sigma_{FTE} = 2.7\text{ m}$, the risk for VAL of 10 m precisely meets the risk requirement of 10^{-6} . Previous works have shown that σ_{FTE} of the multi-copter UAV is a sub-meter level [5]. Based on the conservative value of $\sigma_{FTE} (= 1.5\text{ m})$, the VAL of 10 m for Coverage I is determined.

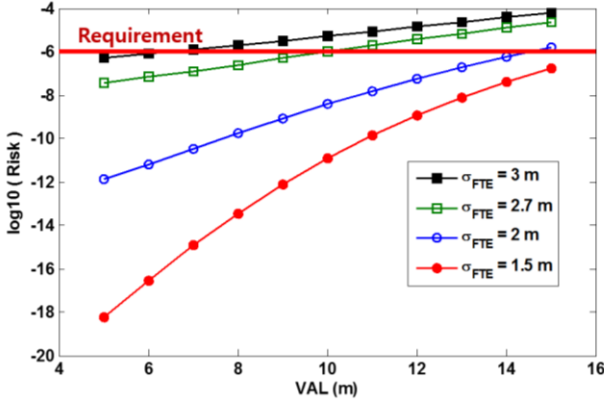


Figure 3. Fault-free risk for Coverage I

3.2 Coverage I and II Tolerable Error Limit

Undetected ionospheric spatial gradients can produce a large user position error. Under general flight conditions, it is very difficult to guarantee that the worst-case undetected ionospheric spatial gradient is bounded by the protection levels computed by the aircraft. To mitigate this threat, GBAS CAT I has implemented the ionospheric geometry screening method. The geometry screening computes the worst-case undetected position error (denoted as MIEV) in the ground system and increases one or more broadcast parameters that affect the airborne protection levels. Consequently, satellite geometries that potentially cause hazardous errors are made unavailable. In this method, the unusable satellite geometries are determined by comparing the MIEV of geometries with a tolerable error limit (TEL) in the vertical direction derived from the obstacle clearance surface for precision approach [6]. The TEL for manned aircraft is derived as a combination of the ideal height of the aircraft in the approach phase, 95% percentile of the fault-free TSE, and the obstacle clearance level near the airport according to (4) [7].

$$TEL(D) = H_{AC}(D) - TSE_{95\%}(D) - H_{OCS}(D) \quad (4)$$

where H_{AC} is the height of the aircraft on an ideal glide path at distance D from the runway threshold, and H_{OCS} is the height of the obstacle clearance surface. $TSE_{95\%}$ is given by

$$TSE_{95\%} = 1.96 \sqrt{\sigma_{NSE,v}^2 + \sigma_{FTE,v}^2}; \quad \sigma_{NSE,v} = \frac{VAL}{k_{FFMD}} \quad (5)$$

For the LAD-GNSS for UAV, the MIEV computed by using the UAV satellite geometry is compared to TEL in real time. The TEL for Coverage I and II of LAD-GNSS is established in a similar manner to Eq. (4), with the exception that H_{AC} is substituted with the low limit of Coverage I and II. The TELs for Coverage I and II are shown in Eq. (6).

$$TEL_{LAD-GNSS} = \begin{cases} 50\text{ ft.} - TSE_{95\%}(d), & \text{if } d < 5\text{ km} \\ 200\text{ ft.} - TSE_{95\%}(d) - H_{OCS}(d), & \text{if } 5\text{ km} < d < 20\text{ km} \end{cases} \quad (6)$$

where d is the distance between the ground subsystem and the UAV.

4. OPTIMAL ALLOCATION

In this section, we describe the method for optimally allocating the integrity/continuity budgets. As mentioned in Section 1, using the known satellite geometry of UAV, the ground subsystem can compute the PLs for all fault scenarios and optimally allocate the integrity and continuity budgets to make the PLs identical for all fault scenarios. As the allocations are optimized with the PLs for each fault mode, the PLs for fault modes that were bounded by using a maximum-allowable error in range (MERR) method should be newly derived.

Figure 4 show the integrity fault tree of the system to which the optimal allocation was applied. The total integrity risk requirement of 2×10^{-7} per 150 s is allocated. The total system integrity requirement remains unchanged from that of the GBAS CAT I. Instead, sub-allocations are adjusted by the inclusion of the optimal allocation. In the re-designed integrity fault tree, the integrity risk of single deformation, low signal level, and all other failures are pre-allocated. This is because that the signal deformation error is canceled out when the ground subsystem and UAV use the receivers that are either the same or very similar. Thus, the small value (1×10^{-8} per 150 s) is allocated to the signal deformation integrity risk. In the case of the low signal level integrity risk, the 1×10^{-8} per 150 s is also pre-allocated because the bound of the low signal level is determined based on the impact on the performance of other monitors, not on MERR. Finally, as all other failures cannot be bounded by MERR or by nominal VPL, the integrity risk, which is the same as that of GBAS CAT I, is allocated.

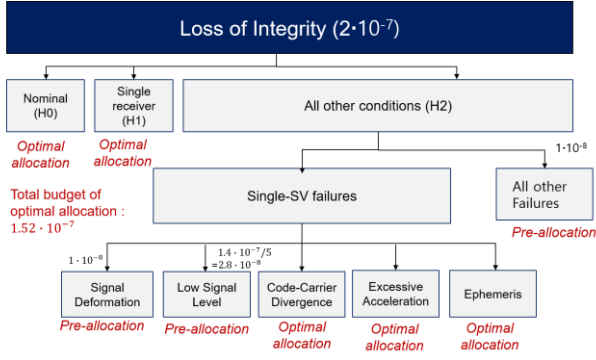


Figure 4. Integrity risk allocation of LAD-GNSS for UAV

4.1 Derivation of Excessive Acceleration Protection Levels

The excessive acceleration (EA) threat is a fault condition in a GNSS satellite that causes the range measurements to excessively accelerate. The most probable causes would be GNSS clock anomalies or ground receiver failures by verifying the consistency of the pseudorange and carrier-phase measurements over the last few epochs. The monitor is designed to detect EA faults by comparing the estimation of the range acceleration error with a threshold [8, 9]. The range acceleration error at epoch k is estimated by

$$acc(k) = \frac{\rho_{c,i}(k+2) - \rho_{c,i}(k) + \rho_{c,i}(k-2)}{(2T)^2} \quad (7)$$

where $\rho_{c,i}$ is the smoothed pseudo-range correction for satellite i , and T is the measurement update time interval. For the LAD-GNSS ground subsystem, this value is 0.5 s. The effective differential ranging error ($\delta\rho_{EA}$) due to an acceleration (acc) for satellite i is:

$$\delta\rho_{EA,i} = 0.5 \cdot \tau \cdot (\tau + T) \cdot acc \quad (8)$$

In this equation, the nominal time delay $\tau = 1.5(ground) + 1(air) = 2.5$ s [9].

The VPL under the EA hypothesis for satellite i consists of two components, as shown in Eq. (9). First, the position error bound (μ_{EA}) is computed based on the bias induced by the satellite fault. Second, the random error term is computed using a sigma-multiplier (k_{HMI}), which considers that monitors miss-detect such faults.

$$VPL_{EA}(i) = \mu_{EA,i} + k_{HMI,EA} \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \quad (9)$$

where k_{HMI} is a multiplier defined to ensure a specified probability of the missed detection, and σ_i is the standard deviation of the differential ranging error for satellite i . Additionally, S_{vert} is the row of the weighted-least-squares projection matrix corresponding to the vertical position state [3]. In this paper, the position error bound (μ_{EA}) for EA is derived by using the minimum detectable error (MDE) concept [3] and Eq. (8).

$$\mu_{EA,i} = S_{vert,i} \cdot 0.5 \cdot \tau \cdot (\tau + T) \cdot (k_{cont,EA} + k_{HMI,EA}) \sigma_{t_{EA}} \quad (10)$$

Here, k_{cont} is a multiplier defined to ensure a specified probability of a fault-free alarm, and $\sigma_{t_{EA}}$ is the standard deviation of the nominal test statistic variation (acc). Considering the model of the test statistics, as in Eq. (7), the standard deviation of statistics is

$$\sigma_{t_{EA}} = \sigma_{noise,PR} \sqrt{1^2 + 2^2 + 1^2} \quad (11)$$

where $\sigma_{noise,PR}$ is the standard deviation of the smoothed pseudo-range noise. The standard deviation of statistics with three reference receivers (RR) and the lowest carrier-to-noise ratio at 32 dB-Hz is $1.76/\sqrt{3}$ cm.

4.2 Derivation of Code-Carrier Divergence Protection Levels

The code-carrier divergence monitor is used to detect ionospheric storms and ensure that the divergence of the code and carrier for any given satellite is sufficiently small. The divergence rate is the estimated change rate of ionosphere delay, I , over time. The input to the divergence rate estimator, z , is the raw code minus the carrier measurement. The discrete time equations for the divergence rate estimator are given as [10]:

$$\begin{aligned} d_1(k) &= \frac{\tau_{d1} - T}{\tau_{d1}} d_1(k-1) + \frac{1}{\tau_{d1}} (z(k) - z(k-1)) \\ Div(k) &= \frac{\tau_{d2} - T}{\tau_{d2}} Div(k-1) + \frac{T}{\tau_{d2}} d_1(k) \end{aligned} \quad (12)$$

where τ_d is the time constant whose values are nominally set at $\tau_{d1} = \tau_{d2} = 30$ s, and T is the output interval of the raw measurements. Given a divergence failure with a CCD rate Div and time onset $t = 0$, a differential ranging error ($\delta\rho_{CCD,i}$) due to a divergence rate for satellite i is [10]:

$$\delta\rho_{CCD,i} = |e_g(t - \max[t_{g,0}, 0]) - e_a(t - \max[t_{a,0}, 0])| \quad (13)$$

where e_g is the ranging error due to Div at ground relative to ground steady state, and e_a is a ranging error due to Div at aircraft relative to ground steady state, $t_{g,0}$, and $t_{a,0}$ is the ground filter start time and airborne filter start time, respectively.

Figure 5 shows the worst-case differential ranging error at Div of 0.004 m/s. Note that prior research [10] suggests that a nominal value of σ_{Div} is 0.004 m/s. The worst case occurs when the ground filter has already converged and the aircraft filter has just begun. For a newly acquired satellite, the pseudo-range is used after two smoothing time constant intervals, or 200 s, to reach steady state. Therefore, the worst differential range error at σ_{Div} of 0.004 m/s is 0.074, as shown in Figure 5. Similar to VPL for EA, the VPL for CCD is derived by using the worst differential range error at σ_{Div} of 0.004 m/s and the MDE concept.

$$VPL_{CCD}(i) = S_{vert,i} \cdot (k_{cont,CCD} + k_{HMI,CCD}) \cdot 0.074 + k_{HMI,CCD} \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \quad (14)$$

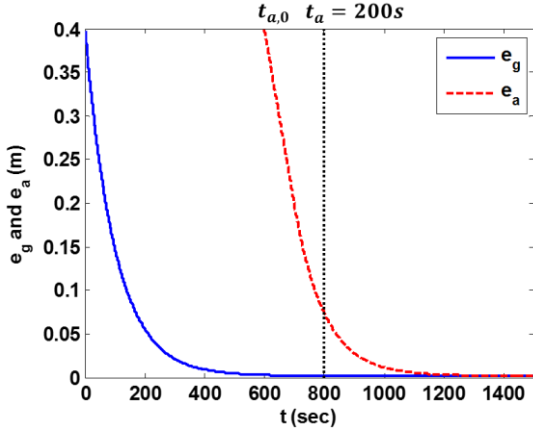


Figure 5. Ranging error due to Div of 0.004 m/s relative to the ground subsystem steady state

4.3 Method of Optimal Allocation

The optimal problem for finding the minima of PLs—which are subject to the constraint that the sum of allocations of each fault mode equals the total risk—is solved by using a Lagrangian function [11]. In this method, all VPLs are expressed as a function of the continuity and integrity risk, as shown in Eq. (15):

$$VPL_i \left(P_{cont,i}, \frac{PHMI_i}{P_{ap,i}} \right) = M_i(P_{cont,i}) + L_i \left(\frac{PHMI_i}{P_{ap,i}} \right) \quad (15)$$

where $P_{cont,i}$ and $PHMI_i$ respectively are continuity and integrity allocated to hypothesis i , and $P_{ap,i}$ is the prior probability of hypothesis i . Once the VPLs are organized into the above expression, a two-dimensional optimization problem can be written as

$$\begin{aligned} & \text{Minimize} \quad \max_i \left[M_i(P_{cont,i}) + L_i \left(\frac{PHMI_i}{P_{ap,i}} \right) \right] \\ & \text{subject to} \quad \sum_{i=0}^{N_{mod}-1} P_{cont,i} = P_{cont} \\ & \text{and} \quad \sum_{i=0}^{N_{mod}-1} PHMI_i = PHMI \end{aligned} \quad (16)$$

where P_{cont} and $PHMI$ are the total continuity and total integrity risk, respectively, and N_{mod} is the number of hypotheses. Using the results of Section 4.1 and 4.2, the VPLs for LAD-GNSS can be expressed as

$$VPL_{H0} = k_{HMI,H0} \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2$$

$$\begin{aligned} VPL_{H1}(j) &= k_{HMI,H1} \sum_{i=1}^n S_{vert,i}^2 \sigma_{H1,i}^2 + |B_{vert,j}| \\ VPL_e(i) &= k_{cont,H0} \cdot \left(\frac{|S_{vert,i}| |x|}{R_i} \sigma_{te} \right) \\ &+ k_{HMI,H0} \left(\frac{|S_{vert,i}| |x|}{R_i} \sigma_{te} + \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \right) \\ VPL_{EA}(i) &= k_{cont,EA,i} \cdot (|S_{vert,i}| \cdot 0.5 \cdot \tau \cdot (\tau + T) \sigma_{tEA}) \\ &+ k_{HMI,H0} \left(|S_{vert,i}| \cdot 0.5 \cdot \tau \cdot (\tau + T) \sigma_{tEA} + \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \right) \\ VPL_{CCD}(i) &= k_{cont,H0} \cdot (|S_{vert,i}| \cdot 0.074) \\ &+ k_{HMI,H0} \cdot \left(|S_{vert,i}| \cdot 0.074 + \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \right) \end{aligned} \quad (17)$$

where σ_{H1} is the standard deviation under a single reference-receiver fault, B_{vert} is the B-value converted to a vertical position error, and x is the displacement vector between the reference station and aircraft antennas. In addition, R_i is the distance (scalar) to satellite i . It is practical at this point to change notations and variables so that the notations are simpler. In Eq. (17), we can change the terms multiplied by continuity, the terms multiplied by integrity, and B-value into θ , ψ , and β , respectively. The problem to be solved is then

$$\begin{aligned} & \text{Minimize} \quad \max_i [k_{HMI,i} \sigma_i + k_{cont,i} \theta_i + \beta_i] \\ & \text{subject to} \quad \sum_{i=0}^{N_{mod}-1} 2a_i Q(-k_{cont,i}) = P_{cont} \\ & \text{and} \quad \sum_{i=0}^{N_{mod}-1} 2P_{ap,i} Q(-k_{HMI,i}) = PHMI \end{aligned} \quad (18)$$

where Q is a unit Gaussian cumulative distribution function, a_0 is zero, and a_i is one. According to [11], by arbitrarily allocating the continuity budget, the optimal problem can be simplified to a one-dimensional search space. In this suboptimal scheme, parameter $k_{cont,i}$ is therefore no longer a variable, and we can define

$$\beta'_i = k_{cont,i} \theta_i + \beta_i \quad (19)$$

The problem to be solved is then

$$\begin{aligned} & \text{Minimize} \quad \max_i [k_{HMI,i} \sigma_i + \beta'_i] \\ & \text{subject to} \quad \sum_{i=0}^{N_{mod}-1} 2P_{ap,i} Q(-k_{HMI,i}) = PHMI \end{aligned} \quad (20)$$

In this paper, we implement the suboptimal scheme because the optimal allocation of the continuity offers only a minor improvement while increasing the complexity of the algorithm and computation load [11].

From the suboptimal scheme described above, as well as the VPLs derived in Sections 4.1 and 4.2, simulations of a VPL under an optimal allocation were conducted. For the

simulations, the continuity risks were allocated to be the same as those of GBAS CAT-I, and the standard RTCA 24 constellation was used to generate visible satellite geometries. The satellite geometries were simulated every 5 min (which gives 288 epochs for one day).

In addition, we assumed that $\sigma_{t_{EA}}$ is 0.99 cm/s^2 at C/N_0 of 32 db-HZ and σ_{Div} is 0.004 m/s . For the standard deviation of ground error (σ_{pr_gnd}), the ground accuracy designator (GAD)-B model with $M = 1$ was used. The latter was used because the effect of mitigating the multipath error is not expected in an environment where the separation distances between the antennas are narrow. The standard deviation of airborne error (σ_{air}) was computed by using the airborne accuracy designator (AAD)-B model for the simulation.

Figure 6 shows the resulting VPLs for Coverage I and II when the UAV operated 5 km apart from the ground subsystem with a constant speed of 15 m/s in Coverage I and 20 km apart from ground subsystem with a constant speed of 45 m/s in Coverage II. In this result, the VPLs for fault-free (H_0), single reference-receiver fault (H_1), ephemeris fault cases, EA-fault cases, and CCD-fault cases were computed by using the integrity risk of the GBAS CAT-I.

As show in Figure 6, the VPL under the optimal allocation of integrity risk, which is shown by the black curve, is slightly below the VPL H_0 . These results show that the VPLs of Coverage I and II are reduced by approximately 8% by using the optimal allocation. The VPL under the optimal allocation of integrity is reasonably well-bounded by VAL of 10 m, which is the operation requirement for Coverage I and II.

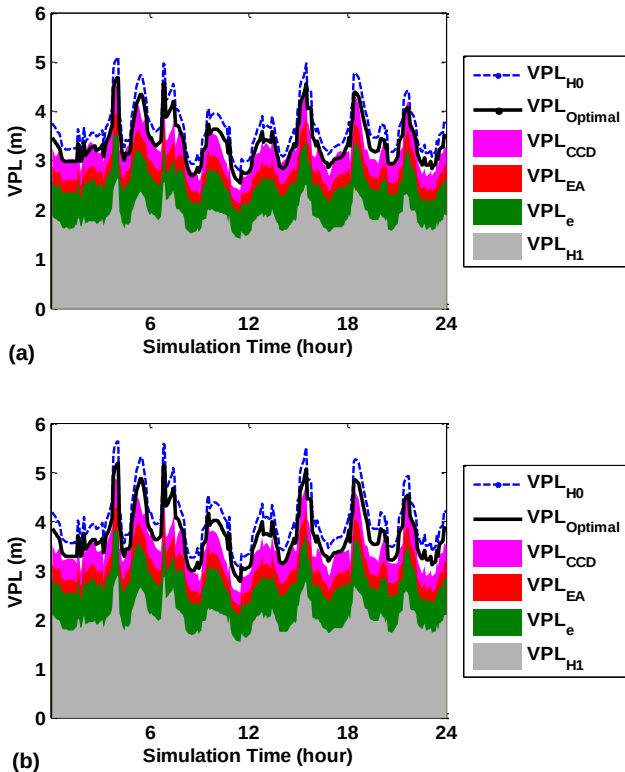


Figure 6. VPLs of LAD-GNSS for UAV operating in (a) Coverage I and (b) Coverage II

5. MITIGATION OF IONOSPHERIC INTEGRITY RISK

To mitigate the integrity risk due to anomalous ionosphere gradients, several monitors, including the ground CCD monitor, DSIGMA, and the simplified geometry screening monitor were implemented in LAD-GNSS. The CCD monitor and DSIGMA could detect and exclude the satellite impacted by large ionospheric gradients using the divergence rate of code-carrier measurements. However, the ionosphere gradients that moved at a slow speed were difficult to detect by both the ground CCD monitor and DSIGMA, which resulted in unacceptable large residual errors in GNSS range signals, even after applying differential corrections. Thus, the simplified geometry screening monitor operated to compute MIEV in real time.

To assess the impacts of anomalous ionosphere gradients on LAD-GNSS users, the worst differential range error was first tabulated using an exhaustive search over combinations of the simulation parameters in the presence of the ground CCD monitor and DSIGMA. Based on the table of the worst differential range errors, the performance of the geometry screening monitor was evaluated according to the geometry simulation.

5.1 Worst-case Differential Range Error Due to Anomalous Ionosphere Gradient

This section describes how we model the worst differential errors in the range domain caused by anomalous ionosphere gradients. In LAD-GNSS, the code-carrier smoothing is processed to mitigate the receiver noise and multipath on both airborne and ground receivers. However, code-carrier divergence due to ionospheric gradients induces bias in the code-carrier smoothing. Such errors in carrier-smoothed code become larger as either a smoothing time or a magnitude of gradient becomes larger. In LAD-GNSS, both the aircraft and ground facility use a time constant of 30 s for the code-carrier smoothing filter in the presence of the anomalous ionosphere gradient.

The resulting differential range errors can be computed by taking the difference between the smoothed ionospheric delay measurements observed from aircraft and the ground facility. Undetected ionosphere-induced differential range errors are then generated for a hypothetical UAV flight in the presence of the modeled ionospheric gradient. The anomalous ionosphere gradient is modeled with several parameters, including the magnitude, width, direction, and relative speed of an ionospheric front with respect to a satellite IPP observed from a ground facility. The moving direction of the UAV and the initial position of both the UAV and ionospheric front are also included in the simulation parameters.

The simulation parameters and ranges are listed in Table 2. For a given set of simulation parameters, the differential range errors and the test statistic of the monitors are computed every epoch as the ionospheric front and UAV advance. When the UAV finally reaches the evaluation point, the differential range error, either

detected or undetected, is recorded. The three different UAV operation scenarios are considered and are summarized in Table 3. The evaluation point is assumed to be 5, 10, and 20 km apart from the ground facility, respectively. The UAV horizontal speed is set at 15 m/s in Coverage I, and 45 m/s in Coverage II, because the speed limit of UAV is constrained to be less than 45 m/s according to Federal Aviation Administration (FAA) regulations.

Table 2. Simulation parameters

Parameter	Min.	Max.	Step Size
Δv (m/s)	-100	350	10
g (mm/km)	200	500	20
W (km)	25	75	25
α (deg)	-90	90	15
β (deg)	-90	90	10

Table 3. Summary of the operating scenario

Scenario	Evaluation Point (km)	Horizontal speed of UAV (m/s)	Coverage
(a)	5	15	I
(b)	10	45	II
(c)	20	45	II

For the monitor threshold, the MDE is used to account for the noise effect. Thus, the MDE values for the ground CCD monitor and DSIGMA are 0.08217 m/s and 2.02 m, respectively. The test statistic of DSIGMA is only generated when the UAV is within the maximum coverage of 20 km.

Figure 7 depicts an example of the results of the worst differential range error in terms of the relative speed of the front. The worst differential range error curves varying with the magnitude of ionosphere gradients from 200 mm/km to 500 mm/km are searched based on the exhaustive simulation over all possible combinations of the ionosphere parameters for each given Δv . The constant model (black line) is proposed to conservatively cover the simulated worst range error curves for all conditions. Since the single-impact scenario is only considered in this analysis, the model of the differential range error does not need to be expressed as a function of Δv . The results of the worst differential range error for the three UAV operation scenarios are summarized in Table 4.

Table 4. Maximum of the worst differential range error for each of three UAV operation scenarios

Operation scenario	(a)	(b)	(c)
Maximum of range error (m)	3	6.5	11.5

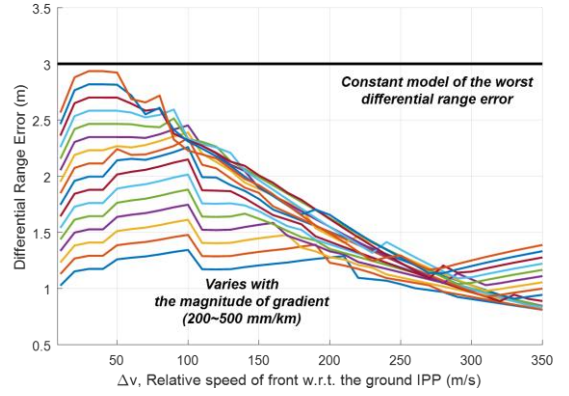


Figure 7. Simulated results of the worst differential range error when the evaluation point is set at 5 km apart from the ground facility and the horizontal speed of UAV is 15 m/s

5.2 Vertical Position Error Due to Ionosphere Gradient

The analytical model of the ionosphere-induced differential range error was introduced in Section 5.1. Using this model, we examine the largest potential errors resulting from the worst case scenarios. To evaluate the performance of the simplified geometry screening monitor, a simulation is conducted to compute the worst-case undetected vertical ionospheric errors by assuming the worst case SV-user geometry and parameter combinations. In this paper, a single-satellite impact is assumed to be the worst case scenario instead of the double-satellite impact mentioned in [12]. This is because the double-impact scenario is a rather conservative assumption for UAV applications.

For a given single satellite i , a vertical ionospheric error is defined as:

$$IEV_i = |S_{vert,i} \varepsilon_i| \quad (21)$$

where $S_{vert,i}$ and ε_i denote the vertical position component of the weighted-least squares projection matrix S and the ionosphere-induced range error for satellite i , respectively. The ionosphere gradient is assumed to move in the worst direction; thus, the resulting Δv is always small. Accordingly, ε_i is always taken as its maximum in the single-impact scenario.

Among IEVs computed for all individual satellites in view, the largest IEV is taken as the MIEV. The MIEV is then compared to the limit of the allowable range error due to anomalous ionosphere gradient, TEL. In this paper, we can derive the TEL using Eqs. (5) and (6). σ_{TSE} is calculated as the square root sum of the fault-free navigation sensor error (NSE) and flight technical error (FTE). σ_{NSE} is derived from the VAL of 10 m, and σ_{FTE} is set at 1.5 m. The obstacle clearance level should be carefully determined because it depends on the environment above which the UAV is actually flying. However, H_{OCS} is provisionally assumed to be 0 m in Coverage I and 15 m (as high as four-floor buildings) in

Coverage II. The resulting TEL for each coverage is 10.8 m and 41.5 m, as shown in Figure 8.

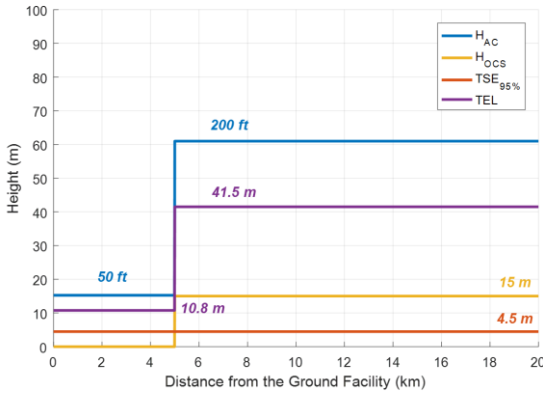


Figure 8. Derivation of TEL for the UAV application

The performance of the geometry screening was evaluated by using the MIEV simulation based on 31 GPS constellations over the course of a day. The satellite geometries were simulated every 5 min (which produced 288 epochs for one day). The ground facility was assumed to be located at Ronald Reagan Washington National Airport, Washington, DC. The MIEVs were computed for both ideal and non-ideal scenarios. “Ideal” means that the UAV loses no satellite in view. However, since the UAV can easily lose satellites on account of low elevation noise or intense scintillation, the non-ideal scenario of a two-satellite outage is also considered. In the non-ideal scenario, the two satellites whose elevation angles are the maximum or minimum are removed from the all-in-view satellites.

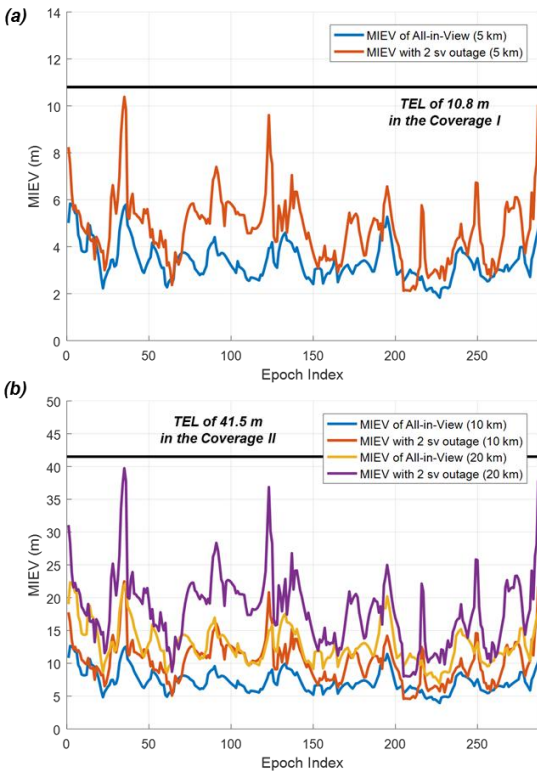


Figure 9. Results of the MIEV simulation for (a) Coverage I and (b) Coverage II

Figure 9(a) shows the result of the MIEV simulation for Coverage I when the UAV operates 5 km apart from the ground facility with a constant speed of 15 m/s. Blue and red curves indicate the MIEV for the ideal and the non-ideal scenarios, respectively. The result shows that the MIEV is well below the TEL of 10.8 m in the ideal scenario, whereas the MIEV for the non-ideal scenario barely satisfies the operation requirement. Sometimes the MIEV of the ideal scenario is even less than the MIEV of the non-ideal scenario. This occurs when the number of tracking satellites is adequately large (e.g., 12 for instance). It implies that the number of satellite is not the only factor contributing to the MIEV.

The MIEV results for Coverage II are shown in Figure 9(b). The UAV is assumed to fly as fast as the speed limit of 45 m/s and is evaluated at two points, which are 10 km and 20 km from the ground facility. When the UAV operates within 10 km, the maximum of MIEV for the non-ideal scenario is increased to approximately 20 m, which sufficiently assures meeting the requirements for Coverage II. However, as the operation distance increases, the MIEV increases. When the UAV is at 20 km, the maximum of the MIEV for the non-ideal condition is slightly below the TEL of 41.5 m. Although the offline simulation results demonstrate that the MIEVs for both Coverage I and II are well below the operation requirements, the MIEV should be monitored in real time in the worst environment case, such as an urban area where it is likely to lose more than two satellites.

6. Conclusion

In this paper, we presented the LAD-GNSS architecture for UAVs that communicate with ground subsystems in real time. LAD-GNSS for UAV was fully analyzed and implemented. It consists of an operation requirement design, integrity monitoring strategies, and optimal allocation of integrity/continuity budgets. The operation coverages that were defined for two regions have respective operating requirements. For Coverage I, VPL was compared to VAL of 10 m, and MIEV was compared to TEL of 10.8 m, to determine if the system met the integrity and continuity requirements for a given operation. The UAV operating Coverage II was expected to meet the VAL of 10 m and TEL of 41.5 m.

In this paper, we derived the PLs for EA and CCD fault cases to implement the optimal allocation of integrity/continuity budgets. We simulated the VPL under an optimal allocation. In the simulation results for a 24-satellite GPS constellation, VPLs of Coverage I and II were reduced by approximately 8% when implementing the optimal allocation method. In addition, we simulated the worst-case undetected vertical ionospheric errors to evaluate the performance of the simplified geometry screening monitor.

The results showed the MIEVs for Coverage I and II were reasonably well-bounded by the TELs, which were pre-determined based on the uncertainty of navigation. The NSE and FTE errors highly depended on the hardware,

environment, and operational conditions and thus needed to be experimentally evaluated under various conditions. To this end, future work will include both the simulation-based analysis and field tests of several types of UAVs in different flight environments and modes.

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